

Recap

DDoS attacks differ depending on their implementation goals

Sophisticated technologies can be deployed to prevent and mitigate DDoS attacks

Another relevant security threat refers to data leakage/breach, that is, the intentional or unintentional disclosure of sensitive data

Unintentional data leakage

General tendency to overshare online and overlook the implications and security risks

Apps & social media decrease the *privacy* of individuals



- Apps with permissions to access (sensitive?) data on our devices with **our consent**
- Location data associated with posts, tweets, photos, ...

Social engineering techniques used to persuade individuals to disclose personal data by stimulating their curiosity and taking advantage of their inexperience and gullibility (e.g., phishing)



Loss/theft of mobile devices or removal media



77

US travel, what changes with social control in the last five years

The United States wants to make it compulsory to declare five years of social media activity in order to obtain an Esta and travel without a visa. The proposal, published by the Department of Homeland Security, also concerns Italian tourists in the Visa Waiver Program

by Angelica Migliorisi
11 December 2025

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

78

Unintentional data leakage: what to do

Importance of raising **awareness** about inadvertent information sharing or oversharing and **educate** individuals

This will avoid the “*crisis of trust*” towards technologies

Every Day Should Be Data Protection Day



JANUARY 28

DATA
PROTECTION
DAY



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

79

Intentional data leakage

Cyber attacks that exploit system/service vulnerabilities or password vulnerabilities to infiltrate a data source and extract sensitive information

Users tend to use the **SAME weak** password across multiple services

Attacks based on brute force (e.g., dictionary attacks, password spraying), credential stuffing

Users seldom change the *default* passwords of their plug-and-play devices (e.g., routers)

Some manufacturers associate *hardcoded* (default) passwords with the (cheap) IoT devices

Lack of password policy enforcement by providers/companies & lack of security awareness of users

Vulnerabilities related to software bugs and misconfigurations (e.g., APIs) are often exploited to compromise privacy

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

80

Passwords: old habits die hard

Generations change, password habits remain

Rank	Password	Count
1	123456	21,627,656
2	admin	21,436,412
3	12345678	8,174,488
4	123456789	5,871,712
5	12345	3,958,777
6	password	3,541,113
7	1qaz!@WSX	2,128,738
8	1234567890	1,418,939
9	Passw0rd	1,228,839
10	admin123	1,467,247
11	1234567	1,486,354
12	123321	1,968,363
13	111111	998,391



Rank	Gen Z	Password
1		12345
2		123456
3		12345678
4		123456789
5		password
6		1234567890
7		skibidi
8		1234567
9		pakistan123
10		assword

Source: NordPass

Simple, short and predictable passwords leave people highly vulnerable to data breaches

Since 2016, billions online credentials stolen

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

81

Data breaches in the news

Mother of all breaches reveals 26 billion records: what we know so far

Updated on: January 29, 2024 10:07 AM

The supermassive leak contains data from numerous previous breaches, comprising an astounding 12 terabytes of information, spanning over a mind-boggling 26 billion records. The leak, which contains LinkedIn, Twitter, Weibo, Tencent, and other platforms' user data, is almost certainly the largest ever discovered.

Big brands caught in 'mother of all breaches'

Twitter, LinkedIn, Adobe among set of 26 billion records.

By Leonard Bernardone on Jan 25 2024 11:48 AM

[Check if your data has been leaked](#)

BRANDS WITH 100M+ LEAKED RECORDS	
BRAND NAME	RECORDS LEAKED
Tencent	1.5B
Weibo	504M
MySpace	360M
Twitter	281M
Wattpad	271M
NetEase	261M
Deezer	258M
LinkedIn	251M
AdultFriendFinder	220M
Zynga	217M
Luxottica	206M
Evite	179M
Zing	164M
Adobe	153M
MyFitnessPal	151M
Canva	143M
JD.com	142M
Badoo	127M
VK	101M
Youku	100M

cybernews

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

82

News coverage

Nike says it is investigating possible data breach

By Nicholas P. Brown and Raphael Satter

January 26, 2026 6:13 PM GMT+1 · Updated January 26, 2026

Penn Data Breach Involves Decades of Student and Alumni Information

The hacker seemed focused on the Ivy League school's admissions preferences.

South Korea blames Coupang data breach on management failure, not sophisticated attack

By Heekyong Yang and Hyunjoon Jin

February 10, 2026 6:16 AM GMT+1 · Updated February 10, 2026

- Coupang data breach affected 33.7 million customers, probe finds
- Former engineer accused of exploiting authentication vulnerabilities

Almost 1 billion Salesforce records stolen, hacker group claims

By Reuters

October 3, 2025 9:35 PM GMT+2 · Updated October 3, 2025

UBS reports a data leak after a cyber attack on a provider

By Reuters

© 2 min read · Published 10:23 AM EDT, Wed June 18, 2025

Hackers leak Qantas data containing 5 million customer records after ransom deadline passes

Gucci, Balenciaga and Alexander McQueen private data ransomed by hackers

15 September 2025

Share

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

83

Pwned websites

Breached websites that have been loaded into Have I Been Pwned



Instagram Data Breach

What Happened

In January 2026, data allegedly scraped via an Instagram API was posted to a popular [hacking forum](#). The dataset contained 17M rows of public Instagram information, including usernames, display names, account IDs, and in some cases, geolocation data. Of these records, 6.2M included an associated email address, and some also contained a phone number. The scraped data appears to be unrelated to password reset requests initiated on the platform, despite coinciding in timeframe. There is no evidence that passwords or other sensitive data were compromised.

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

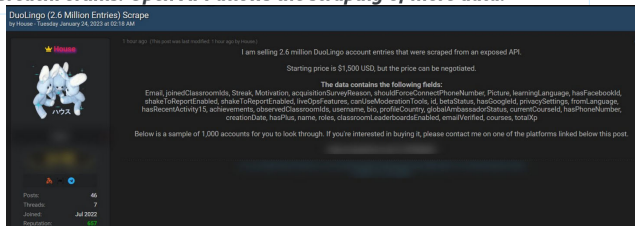
84

Data leaks due software bugs

Hackers exposed 2.6 million Duolingo users, more available for scraping

Updated on: November 15, 2023 12:53 PM

Duolingo, the popular language learning app, has had some of its users exposed online. The scraped data of 2.6 million people, which was on sale in January, is now available on the cybercrime marketplace BreachForums. Open API allows the scraping of more data.



OpenAI CEO admits a bug allowed some ChatGPT users to see others' conversation titles

PUBLISHED THU, MAR 23 2023-9:58 AM EDT | UPDATED MON, APR 17 2023-2:00 AM EDT

 **Ryan Browne**
@RYAN_BROWNE

© M. C. Calzarossa 2026

A Nasty Safari Bug Could Leak Your Browsing Data And Google Account Information

By [Lee Mathews](#), Former Contributor. Observing, pondering, and writing about tech. Generally in that order.

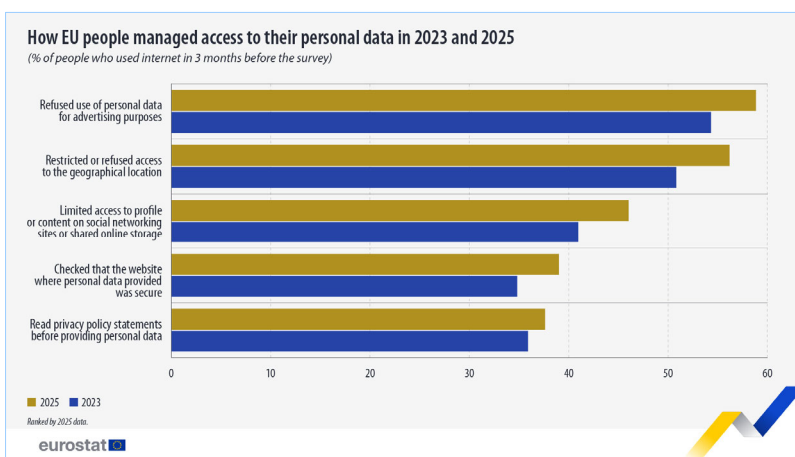
Jan 11, 2022, 03:40pm EST

OpenAI CEO Sam Altman said the company fixed a vulnerability in its AI chatbot ChatGPT that meant some users were able to see the titles of chats had by others.

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

Users and personal data: Eurostat survey

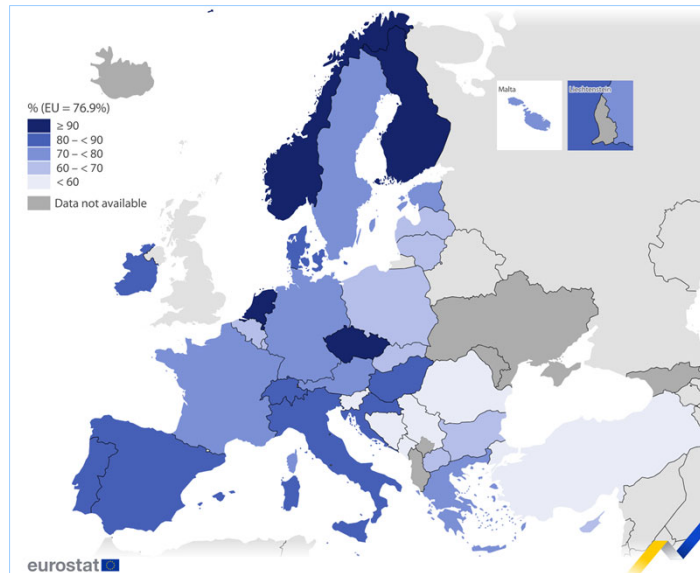


In 2025 76.9% of EU internet users protected their data online by taking steps to manage access to their personal data

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

Data protection by country

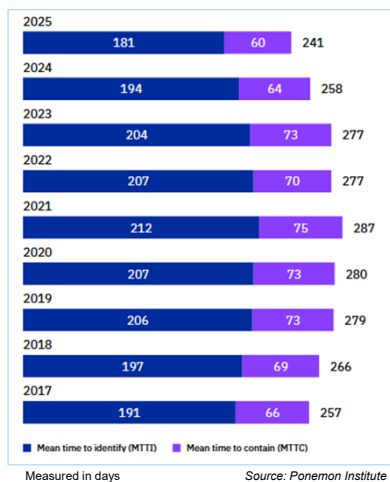


© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

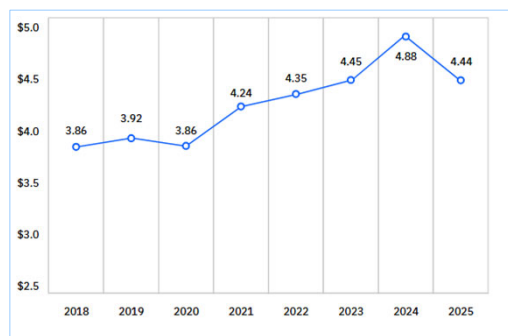
87

Time to identify/contain a data breach & cost



Source: Ponemon Institute

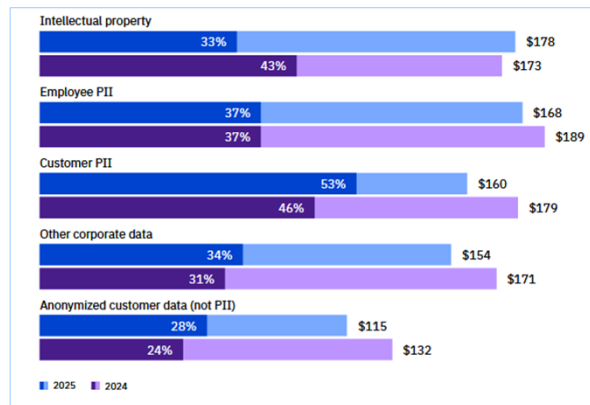
© M. C. Calzarossa 2026



Enterprise Digital Infrastructure

88

Type of data compromised & cost



Measured in million USD

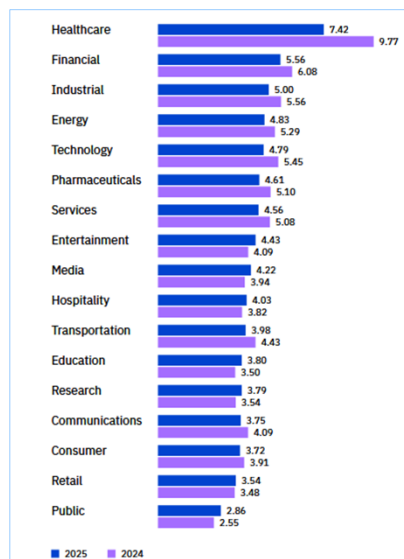
Source: Ponemon Institute

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

89

Cost of a data breach by vertical



Measured in million USD

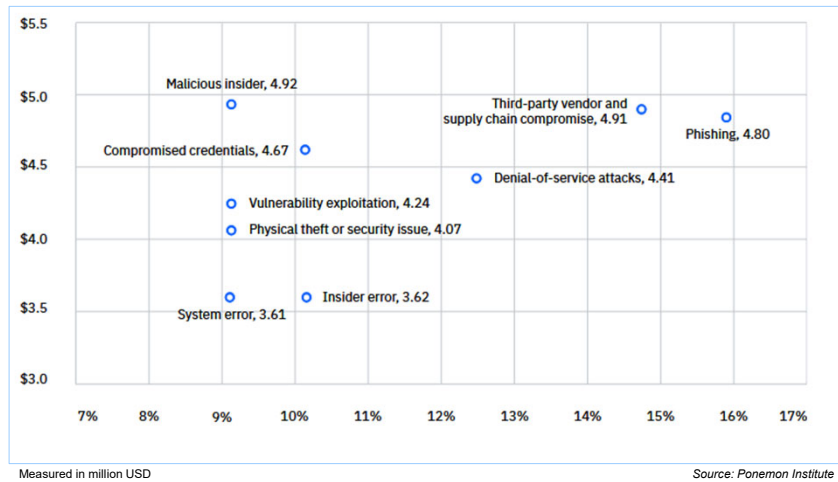
Source: Ponemon Institute

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

90

Cost & frequency by attack vector



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

91

How to protect data & infrastructure

Two different perspectives:

1. User's perspective
2. Provider's perspective

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

92

User's perspective

Password usage

Passwords must be *strong* and *different* across services (e.g., password managers)

No *default* passwords on any device

Data *oversharing* must be avoided

Firmware, operating systems and applications must be timely updated

Before any click, links (e.g., URLs, attachments) must be carefully checked

Anti-virus tools should be deployed

Whenever possible, VPNs should be used